

EMILY HASTINGS

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EDUCATION

University of Illinois, Urbana-Champaign, IL

Ph.D. in Computer Science

2016-2023

M.S. in Computer Science

2016-2019

Specialization: Human-Computer Interaction

Advisors: Brian Bailey, Karrie Karahalios

Research Interests: team formation, CS education, algorithm awareness, crowdsourcing

Courses included: user interface design, human-computer interaction, experimental methods, educational technology, social media and signals, models of cognitive processes, data mining, educational game design

Knox College, Galesburg, IL

B.A. in Computer Science, *summa cum laude*

2012-2016

Independent Minor: Renaissance and Medieval Studies

Courses included: data structures, hardware organization, information management, algorithm design, graphics, parallel programming, software engineering, networking, operating systems, artificial intelligence

RESEARCH EXPERIENCE

University of Wisconsin-Eau Claire, Eau Claire, WI

Assistant Professor

Fall 2023 – Present

Conducting research in the areas of human-computer interaction, e-textiles, and computer science education.

University of Illinois, Urbana-Champaign, IL

Research Assistant

Fall 2016 – Spring 2023

Advisor: Brian Bailey

Worked individually and with a team to investigate issues concerning the use of algorithmic team formation tools.

National Institute of Standards and Technology, Gaithersburg, MD

Guest Researcher/GMSE Fellow

Summer 2018 – 2020

Advisors: Michael Brundage, Rachael Sexton

Worked with Knowledge Extraction Application team in Engineering Laboratory toward quantifying human skill level from historical data and improving team formation for maintenance.

University of Illinois, Urbana-Champaign, IL

Beyond the Black Box Research Team Member

2018 – 2019

Advisors: Karrie Karahalios (UIUC), Christian Sandvig (UMich)

Worked with a team across multiple universities to conduct a large-scale study on algorithmic literacy and awareness.

Knox College, Galesburg, IL

Research Assistant

Summer 2015

Advisor: Jaime Spacco

Worked with a team to develop Knoxcraft (<https://github.com/knoxcraft>), a system that allows students to use Java/Python code to build structures in the game Minecraft.

Knox College, Galesburg, IL

Research Assistant

Summer 2014

Advisor: David Bunde

Worked with a team to develop materials to help teach parallel programming at Knox and other institutions.

Knox College, Galesburg, IL

Research Assistant

Summer 2013

Advisor: David Bunde

Worked with a team to investigate task mapping and cabling methods for the Dragonfly interconnect topology.

TEACHING EXPERIENCE

University of Wisconsin-Eau Claire

Assistant Professor

2023-Present

Responsible for courses CS 140 Introduction to Computer Science; CS 145 Programming for New Programmers; CS 146 The Big Picture in Computer Science; CS 335 Algorithms; AI 350 Human-Computer Interaction; and HNRS 200 Artificial Intelligence, Algorithms, and Society.

University of Illinois

Teaching Assistant for “User Interface Design”

2021-2022

Ran design studio sections, gave feedback on and evaluated course projects, developed exams, and held office hours. Ranked as Excellent by Students (evaluation: 4.49/5).

Certificate in Foundations of Teaching

2021

Participated in eight hours of teaching development workshops; had an observation of, and reflected on, my teaching; explored literature on teaching; observed an experienced instructor; and wrote a teaching philosophy statement.

Knox College

Teaching Assistant for “Introduction to Computer Science” and “Program Design and Methodology”

2014-2016

Assisted professors during lab sessions, graded homework, lab assignments, and quizzes, and held office hours.

Teaching Assistant in the Costume Shop

2013-2014

Built garments for college theatrical shows, mentored students on individual projects, and presented costume research to classes.

STUDENT RESEARCH SUPERVISED

Ethan Behrendt and Hunter McDaniel

2025-2026

Using E-Textiles for Undergraduate Learning

Funded by ORSP. Poster presented at CERCA 2026. Paper presented at MICS 2026.

Sydney Hoffman (with Dr. Jennine Fox)

2025

Consumer Perceptions of AI-Generated vs. Human-Generated Marketing: Insights and Best Practices

Honors thesis (first from the UWEC College of Business).

Ethan Behrendt

2024-2025

A Review of E-Textiles in Learning Environments

Funded by ORSP. Poster presented at CERCA 2025. Paper presented at ACM C&C 2025.

Jared B. Jones (with Drs. Jim Seliya and Ben Fine)

2024-2025

An Overview of Gender and Political Biases in LLMs

2nd place in EIT 2024 student poster contest. Paper presented at MICS 2025.

PUBLICATIONS AND PAPERS

Ethan R. Behrendt, Hunter M. McDaniel, and **Emily M. Hastings**. 2026. An E-Textiles Workshop for Undergraduate Learning. In *Proceedings of the Midwest Instruction and Computing Symposium 2026*.

Ethan Behrendt and **Emily M. Hastings**. 2025. A Review of E-Textiles in Learning Environments. In *Proceedings of the 2025 Conference on Creativity and Cognition (C&C '25)*. Association for Computing Machinery, New York, NY, USA, 508–511. <https://doi.org/10.1145/3698061.3733252>

Jared B. Jones, Naeem Seliya, **Emily M. Hastings**, and Benjamin T. Fine. 2025. An Overview of Gender and Political Biases in LLMs. In *Proceedings of the Midwest Instruction and Computing Symposium 2025*.

Emily M. Hastings, Vidushi Ojha, Benedict V. Austriaco, Karrie Karahalios, and Brian P. Bailey. 2023. Composing Team Compositions: An Examination of Instructors' Current Algorithmic Team Formation Practices. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW2, Article 305 (October 2023), 24 pages. <https://doi.org/10.1145/3610096>

Emily M. Hastings, Sneha R. Krishna Kumaran, Karrie Karahalios, and Brian P. Bailey. 2022. A Learner-Centered Technique for Collectively Configuring Inputs for an Algorithmic Team Formation Tool. In *Proceedings of the 53rd ACM Technical Symposium on Computer Science Education V. 1 (SIGCSE 2022)*. Association for Computing Machinery, New York, NY, USA, 969–975. DOI:<https://doi.org/10.1145/3478431.3499331>.

Reslan, M., **Hastings, E.**, Brundage, M. P., & Sexton, T. (2021). A Data-Driven Framework for Team Formation for Maintenance Tasks. *IJPHM*, 12, 003.

Emily M. Hastings, Albatool Alamri, Andrew Kuznetsov, Christine Pisarczyk, Karrie Karahalios, Darko Marinov, and Brian P. Bailey. 2020. LIFT: Integrating Stakeholder Voices into Algorithmic Team Formation. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376797>

Hastings, E., Sexton, T., Brundage, M. P., & Hodkiewicz, M. (2019). Agreement Behavior of Isolated Annotators for Maintenance Work-Order Data Mining. *Proceedings of the Annual Conference of the PHM Society*, 11(1). <https://doi.org/10.36001/phmconf.2019.v11i1.791>

Emily M. Hastings, Farnaz Jahanbakhsh, Karrie Karahalios, Darko Marinov, and Brian P. Bailey. 2018. Structure or Nurture? The Effects of Team-Building Activities and Team Composition on Team Outcomes. *Proc. ACM Hum.-Comput. Interact.* 2, CSCW, Article 68 (November 2018), 21 pages. <https://doi.org/10.1145/3274337>

E. Hastings, D. Rincon-Cruz, M. Spehlmann, S. Meyers, A. Xu, D. P. Bunde, and V. J. Leung, “Comparing global link arrangements for dragonfly networks,” in *2015 IEEE International Conference on Cluster Computing*, Sept 2015, pp. 361–370.

PRESENTATIONS AND POSTERS

<i>More Sustainable AI Use and Development: Software Side</i> UWEC AI and Sustainability Symposium	2026
<i>Supporting Instructor Decisions on Algorithmic Team Formation through Integrating Stakeholder Voices</i> UWEC Mathematics Colloquium	2025
<i>A Review of E-Textiles in Learning Environments</i> (poster) UWEC Celebration of Scholarship	2025
<i>A Review of E-Textiles in Learning Environments</i> ACM Conference on Creativity and Cognition	2025
<i>Artificial Intelligence in Research</i> (panel discussion) UWEC Faculty/Academic Staff Forum Fall 2024 Series	2024
<i>Composing Team Compositions: An Examination of Instructors' Current Algorithmic Team Formation Practices</i> ACM Conference on Computer-Supported Cooperative Work	2023
<i>Supporting Instructor Decisions on Algorithmic Team Formation through Integrating Stakeholder Voices</i> Bradley University CSIS Colloquium	2022
<i>Supporting Instructor Decisions on Algorithmic Team Formation through Integrating Stakeholder Voices</i> University of Wisconsin-Eau Claire CS Colloquium	2022
<i>Supporting Instructor Decisions on Algorithmic Team Formation through Integrating Stakeholder Voices</i> Lawrence University MSCS Colloquium	2022
<i>Supporting Instructor Decisions on Algorithmic Team Formation through Integrating Stakeholder Voices</i> St. Olaf College MSCS Colloquium	2022
<i>A Learner-Centered Technique for Collectively Configuring Inputs for an Algorithmic Team Formation Tool</i> ACM Technical Symposium on Computer Science Education	2022

<i>LIFT: Integrating Stakeholder Voices into Algorithmic Team Formation</i>	
Knox College Computer Science Colloquium	2021
<i>LIFT: Integrating Stakeholder Voices into Algorithmic Team Formation</i> (poster)	
EECS Rising Stars	2020
<i>LIFT: Integrating Stakeholder Voices into Algorithmic Team Formation</i>	
ACM CHI Conference on Human Factors in Computing Systems	2020
<i>Structure or Nurture? The Effects of Team-Building Activities and Team Composition on Team Outcomes</i>	
ACM Conference on Computer-Supported Cooperative Work	2018
<i>The History and Construction of Elizabethan English Costume</i>	
Knox College Presentation of Independent Study Research	2016
<i>Knoxcraft: Teaching Introductory Programming with Minecraft</i> (poster)	
Knox College Horizons Celebration of Student Research	2016
<i>Knoxcraft: Teaching Introductory Programming with Minecraft</i>	
Knox College Summer Science Seminar Series	2015
<i>Adventures in Parallel Programming</i> (poster)	
Knox College Horizons Celebration of Student Research	2015
<i>Adventures in Parallel Programming</i> (Best Student Seminar Award)	
Knox College Summer Science Seminar Series	2014
<i>The History and Construction of Elizabethan English Costume</i> (poster)	
Knox College Horizons Celebration of Student Research	2014
<i>Dragonfly Interconnect Topology</i> (poster)	
Knox College Horizons Celebration of Student Research	2014
<i>Dragonfly Interconnect Topology</i>	
Knox College Summer Science Seminar Series	2013

AWARDS AND HONORS

Karlgaard Faculty Excellence Award, <i>UWEC</i>	2025
List of Teachers Ranked as Excellent, <i>University of Illinois</i>	2021, 2022
Tau Beta Pi, <i>University of Illinois</i>	2021
Invited participant to EECS Rising Stars 2020, <i>UC Berkeley</i>	2020
Mavis Future Faculty Fellowship, <i>University of Illinois</i>	2020-2021
Graduate Measurement Science and Engineering Fellowship, <i>NIST/GFSD</i>	2018-2020

Phi Beta Kappa, <i>Knox College</i>	2016
E. Inman Fox Prize, <i>Knox College</i>	2016
Paul's Prize in Computer Science, <i>Knox College</i>	2016
Howard A. Wilson Prize in Literary Criticism (2nd Place), <i>Knox College</i>	2016
ASSET Scholar, <i>Knox College</i>	2015-2016
Ron Asplund Memorial Research Award, <i>Knox College</i>	2014
National Merit Scholar, <i>Knox College</i>	2012-2016

SKILLS

Microsoft Office, Google App Suite, Windows, IntelliJ, Eclipse, Github
 Programming languages (high proficiency): Java
 Programming languages (some experience): Python, C/C#, SQL, HTML/CSS, Javascript/JQuery, PHP, Android development, game development in Unity
 Knowledge of research methodologies
 Knowledge of statistical analysis techniques, R
 Learning management systems: Canvas, Blackboard Learn, Moodle
 Writing and presenting reports
 English (native language)
 Elementary proficiency in French and Latin

SERVICE AND LEADERSHIP

<i>Midwest Instruction and Computing Symposium</i> Session Chair for MICS 2026	2026
<i>University of Wisconsin-Eau Claire</i> Honors College Affiliate	2026-2027
CS Department Curriculum and Scholarship Committees	2026-present
Website Editor for CS Department	2025-present
Brains & Bytes: Adventures of Search and Sort Games Outreach event at The Children's Museum of Eau Claire sponsored by the CS department. Assisted with planning and facilitating activities during the event.	2025
Website Editor for CS Department	2025-present
Faculty Advisor for UWEC SACM	2024-present

Sonia Kovalevsky Day 2024 Committee member and volunteer coordinator. Assisted with planning and facilitating activities during the event.	2024
<i>Institute of Electrical and Electronics Engineers</i> Reviewer for IEEE EIT	2024-present
Registration Co-chair, Session Chair for IEEE EIT 2024	2023-2024
<i>Behaviour & Information Technology (Journal)</i> Reviewer	2020
<i>Association for Computing Machinery</i> Reviewer for ACM CHI	2020-present
Reviewer for ACM CSCW	2019-present
<i>University of Illinois</i> CS STARS Research Mentor Supervised undergraduate student research project.	2022
Grad Academy Small Group Leader Co-led a training session for new CS teaching assistants.	2021
Girls Who Code Facilitator Assisted students during weekly club meetings.	2017
Engineers Volunteering in STEM Education (ENVISION) Led school age children in STEM-related activities.	2016-2017
<i>Knox College</i> Teaching Assistant for Knox College 4 Kids Assisted teachers for three summers teaching knitting, crochet, weaving, French, and Harry Potter classes to school-age children.	2011-2013

MEMBERSHIPS

Association for Computing Machinery (ACM)
 ACM Special Interest Group on Computer-Human Interaction
 Phi Beta Kappa Honor Society
 Tau Beta Pi Honor Society